

IA: MACHINE LEARNING FOUNDATIONS

SESSION 6 – Overfitting and Regularization

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AGENDA

1. Theory of generalization
2. Approximation-Generalization Tradeoff
3. Overfitting
4. Cross-Validation
5. Regularization
6. Evaluation Metrics



1. THEORY OF GENERALIZATION

What must be your real objective in this course?



1. THEORY OF GENERALIZATION

What must be your real objective in this course?

FAIL

PASS THE EXAM

LEARN





1. THEORY OF GENERALIZATION

E_{out}

Measures how well our training on D has generalized to data not seen before.

I we want to estimate its value with a sample, these points must be “fresh” test points not used during training.

E_{in}

Based on data points that have been used for training.

It measures the training performance.

*** NOTE:** more details about this in the book Learning from Data.

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2. APPROXIMATION-GENERALIZATION TRADEOFF

2.1. BIAS AND VARIANCE

Bias

Bias is the difference between the average prediction of our model and the correct value which we are trying to predict. It is also known as **Bias Error** or **Error due to Bias**.

Model with **high bias pays very little attention to the training data** and oversimplifies the model. It always leads to high error on training and test data.

Low Bias models: k-Nearest Neighbors (k=1), Decision Trees and Support Vector Machines.
High Bias models: Linear Regression and Logistic Regression.

Variance

Variance is the variability of model prediction for a given data point or a value which tells us spread of our data. Variance is the **amount that the prediction will change if different training data sets were used**.

Model with high variance pays a lot of attention to training data and does not generalize on the data which it hasn't seen before. As a result, such models perform very well on training data but has high error rates on test data.

Low Variance models: Linear Regression and Logistic Regression.
High Variance models: k-Nearest Neighbors (k=1), Decision Trees and Support Vector Machines.

2. APPROXIMATION-GENERALIZATION TRADEOFF

2.1. BIAS AND VARIANCE

Problems associated with Bias-Variance

High Bias - Low Variance (*Underfitting*): Predictions are consistent, but inaccurate on average. This can happen when the model uses very few parameters and is unable to capture the underlying pattern of the data.

High Bias - High Variance: Predictions are inconsistent and inaccurate on average.

Low Bias - Low Variance: It is an ideal model.

Low Bias - High Variance (*Overfitting*): Predictions are inconsistent and accurate on average. This can happen when the model uses a large number of parameters and when our model captures the noise along with the underlying pattern in data.

2. APPROXIMATION-GENERALIZATION TRADEOFF

2.1. BIAS AND VARIANCE

Identify High Variance

- Low training error (lower than acceptable test error)
- High test error (higher than acceptable test error)

Identify High Bias

- High training error (higher than acceptable test error)
- Test error is almost same as training error

Reduce High Variance

- Reduce input features(because you are overfitting)
- Use less complex model
- Include more training data
- Increase Regularization term

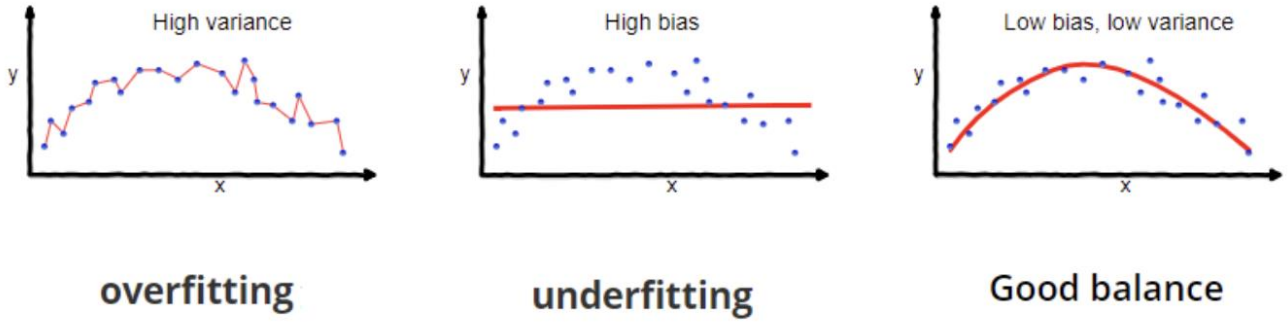
Reduce High Bias

- Use more complex model (Ex: add polynomial features)
- Increase input features
- Decrease Regularization term

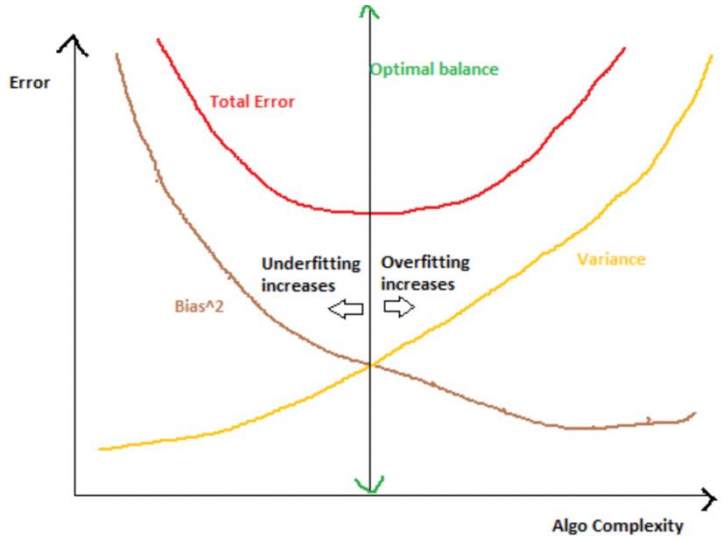
2. APPROXIMATION-GENERALIZATION TRADEOFF

2.1. BIAS AND VARIANCE

- If our model is too simple and has very few parameters, then it may have high bias and low variance.
- On the other hand, if our model has large number of parameters then it's going to have high variance and low bias.



$$Err(x) = \text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$$



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6. Evaluation Metrics



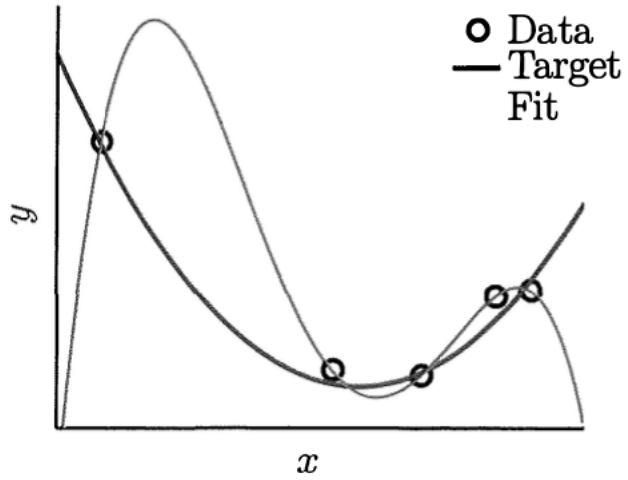
3. OVERFITTING

- Overfitting literally means “Fitting the data more than is warranted”.

$\downarrow\downarrow E_{in}$ $\uparrow\uparrow E_{out}$

- When does overfitting occur?

E_{in} alone is no longer a good guide for learning



Overfitting goes beyond bad *generalization*.

It applies to the process of picking a hypothesis with lower and lower E_{in} resulting in higher and higher E_{out}.

3. OVERFITTING

- In practice, there are some factors that affects overfitting:

Number of data points	↑	Overfitting	↓
Noise	↑	Overfitting	↑
Target complexity	↑	Overfitting	↑

- Resampling methods are the most common techniques to avoid overfitting

- **Underfitting** refers to a model that can neither model the training data nor generalize to new data.
- An **underfit machine learning model** is not a suitable model and will be obvious as it will have poor performance on the training data.
- **Underfitting** is often not discussed as it is easy to detect given a good performance metric. The remedy is to move on and try alternate machine learning algorithms. Nevertheless, it does provide a good contrast to the problem of overfitting.

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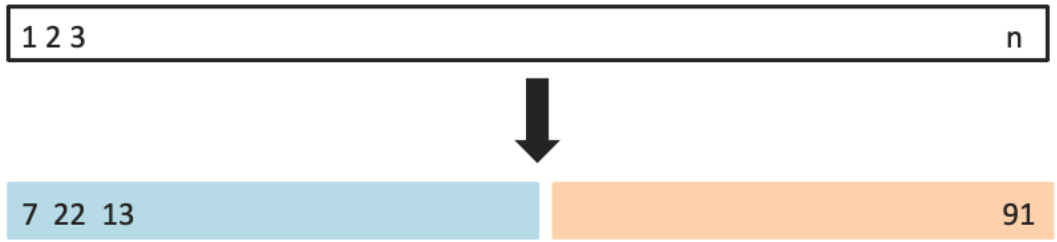
4. CROSS VALIDATION

- Resampling methods involve fitting the same statistical method multiple times using different subsets of the training data.
- Repeatedly drawing samples from a training set and refitting a model of interest on each sample in order to obtain additional information about the fitted model.
- There are two commonly used resampling methods:
  - Cross Validation
  - Bootstrap
- Computationally expensive
- It's a common mistake not to have a representative test set → Training set is used to estimate the test error.

4. CROSS VALIDATION

4.1. TEST SET

- This is a very simple strategy that involves randomly dividing the available set of observations into two parts, a training set and a test set or hold-out set.



- The model is fit on the training set, and the fitted model is used to predict the responses for the observations in the test set.

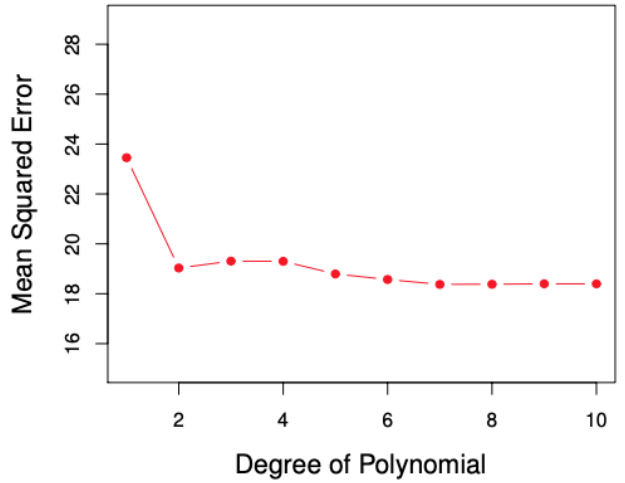
- The resulting test set error rate provides an estimate of the test error rate.

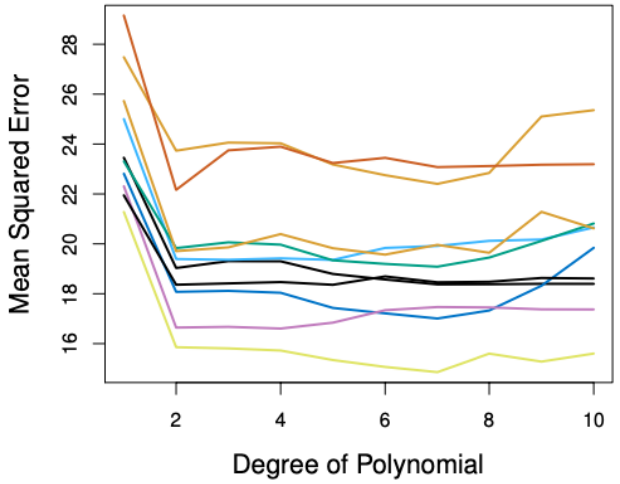
4. CROSS VALIDATION

4.1. TEST SET

This approach is simple, but:

- The test estimate of the test error rate can be highly variable, depending on precisely which observations are included in the training set and which observations are included in the test set.
- Only a subset of the observations are used to fit the model. Since statistical methods tend to perform worse when trained on fewer observations, this suggests that the test set error rate may tend to *overestimate* the test error rate for the model fit on the entire data set.

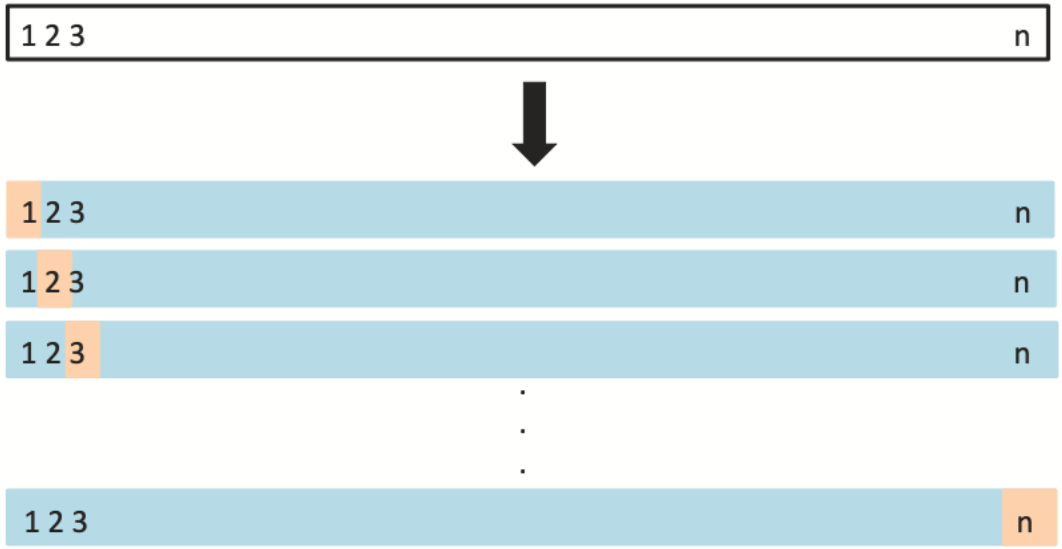




4. CROSS VALIDATION

4.2. LEAVE-ONE-OUT CROSS-VALIDATION

- This technique also involves splitting the training observations into two parts. However, we use a single sample for test and the rest for training:



- Since computing the error over a single observation gives a poor estimate, the final test error rate is the average of the individual N errors.

4. CROSS VALIDATION

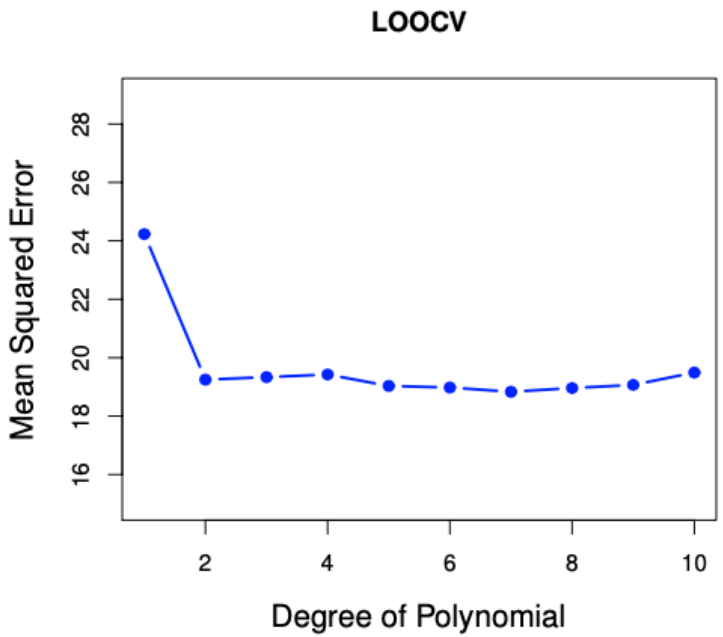
4.2. LEAVE-ONE-OUT CROSS-VALIDATION

LOOCV has a couple of major advantages over the test set approach:

- It has far less bias*. Most of the samples are used to train.
- Performing LOOCV multiple times will always yield the same results. There is no randomness in the training/test set splits.

However:

- It is very time consuming.

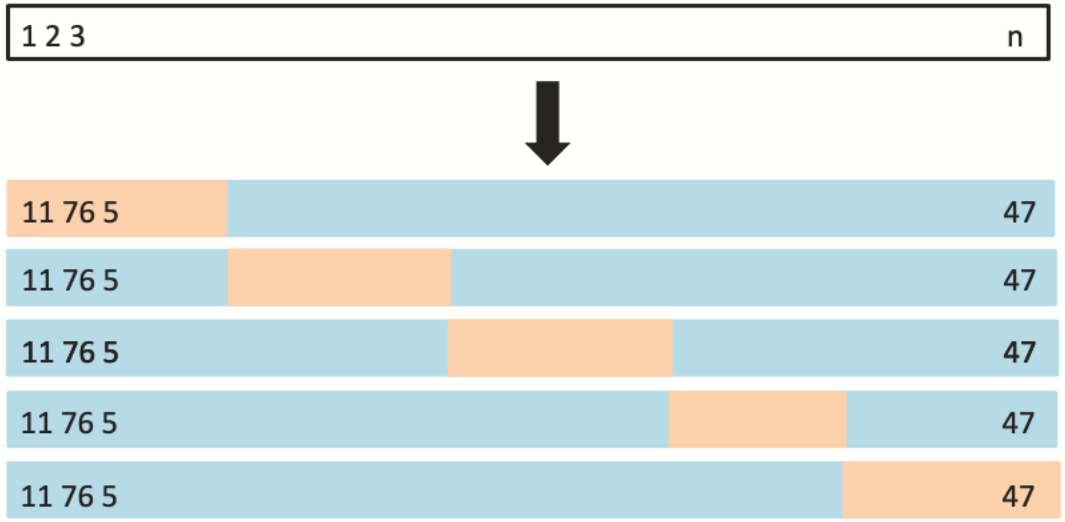


*NOTE: Model with high bias pays very little attention to the training data and oversimplifies the model. It always leads to high error on training and test data.

4. CROSS VALIDATION

4.3. K-FOLD CROSS-VALIDATION

- This approach involves randomly dividing the set of observations into *k* groups, or *folds*, of approximately equal size.



- As in LOOCV, the final test error rate is the average of the individual *k* errors. For example, for a regression problem:

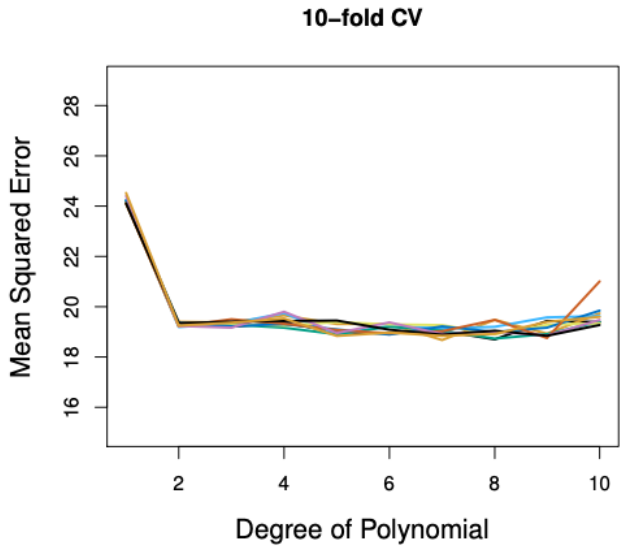
$$\mathrm{CV}_{(k)} = \frac{1}{k}\sum_{i=1}^{k}\mathrm{MSE}_i$$

4. CROSS VALIDATION

4.3. K-FOLD CROSS-VALIDATION

What is the advantage of using $k = 5$ or $k = 10$ rather than $k = n$?

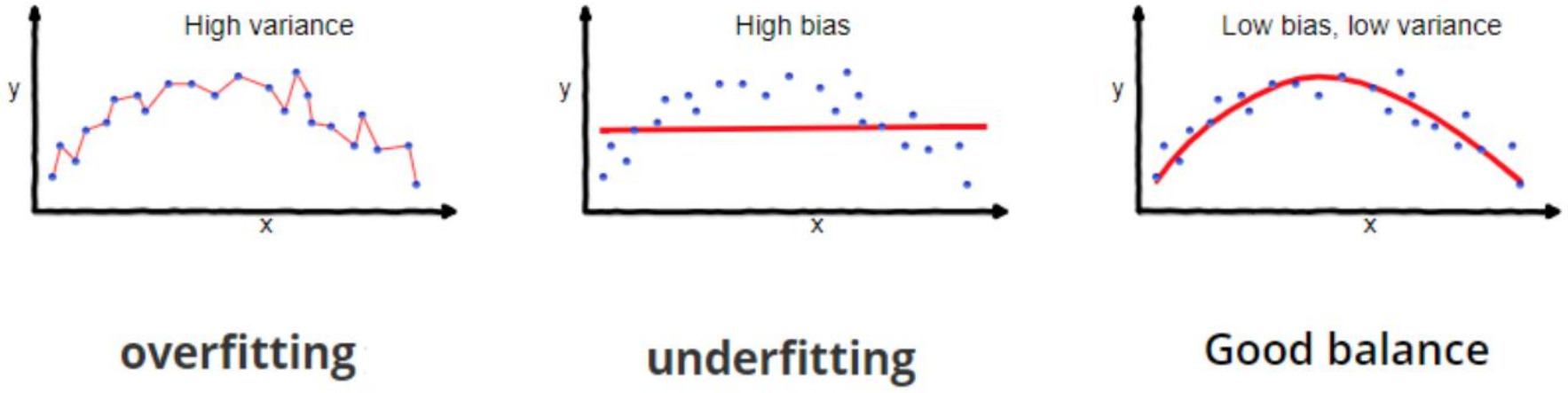
- Computational, especially in complex models.
- Bias-variance trade-off.



4. CROSS VALIDATION

4.4. BIAS-VARIANCE TRADE-OFF FOR K-FOLD CROSS-VALIDATION

- Recap of bias-variance trade-off meaning:



4. CROSS VALIDATION

4.4. BIAS-VARIANCE TRADE-OFF FOR K-FOLD CROSS-VALIDATION

- In terms of BIAS:
  - Train-Test Split overestimates the error rate (high bias) → Training is just an small portion of observations
  - LOOCV will give approximately unbiased estimates → Training set contains *n–1* observations
  - K-Fold give an intermediate level of bias → approximately $\frac{(k-1)n}{k}$ observations

What about the variance?

In terms of BIAS, LOOCV is to be preferred to k-fold CV

4. CROSS VALIDATION

4.4. BIAS-VARIANCE TRADE-OFF FOR K-FOLD CROSS-VALIDATION

- In terms of VARIANCE:
  - It is clear that, in general, Train-test Split is the worst option
  - With LOOCV we average N almost identical models → Higher variance
  - With K-Fold CV, we average *k* less correlated models since the overlap between the training sets in is smaller → Lower variance

Generally speaking, K-Fold CV with k=5 or k=10 have been shown empirically to yield test error rate estimates that suffer neither from excessively high bias nor from very high variance

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5. REGULARIZATION

5.1. RIDGE REGRESSION

- Apart from the subset selection, there exists another common technique to improve models performance: regularization (*shrinkage*)

- These methods use all predictors and regularize the coefficient estimates, or equivalently, shrink the coefficient towards zero.

- Ridge Regression is very similar to least squares, except that the coefficients are estimated by minimizing a slightly different quantity:

$$\text{RSS} + \lambda \underbrace{\sum_{j=1}^{p} \beta_j^2}$$

$\lambda \geq 0$ is a tunning parameter

Shrinkage Penalty → ℓ_2 penalty

5. REGULARIZATION

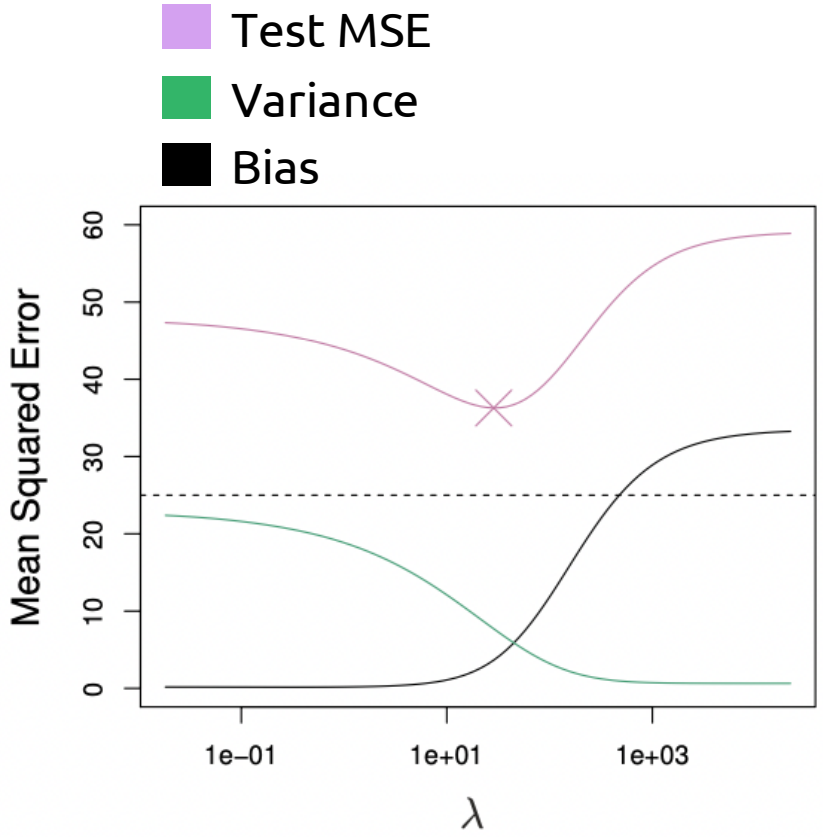
5.1. RIDGE REGRESSION

- Why does Ridge Regression improve over Least Squares?

  Bias-Variance Trade-Off

  - The test MSE is closely related to the variance plus the squared bias
  - For high values of λ the shrinkage on the coefficients causes the model to be significantly underestimated
  - The mínimum MSE is achieved for $\lambda = 30$

- In general, in situations where the relationship between the response and the predictors is close to linear, the least squares estimates will have low bias but may have high variance.



5. REGULARIZATION

5.2. LASSO

- Ridge regression does have one obvious disadvantage: it includes all *p* predictors in the final model.
- AS we have seen before, this may not be the best solution.
- Lasso overcome this diasvantage by minimizing:

$$\text{RSS} + \lambda \underbrace{\sum_{j=1}^{p} |\beta_j|}_{\ell_1 \text{ Penalty}}$$

The ℓ_1 penalty has the effect of forcing some of the coefficient estimates to be exactly equal to zero when the tuning parameter λ is sufficiently large $\longrightarrow$ *variable selection*

- As in Ridge Regression, selection of λ is critical.

5. REGULARIZATION

5.2. LASSO

- It is clear that the Lasso has a major advantage over Ridge Regression, in that it produces simpler and more interpretable models that involve only a subset of the predictors.

- **However, which method leads to better prediction accuracy?**

5. REGULARIZATION

5.2. LASSO

- It is clear that the Lasso has a major advantage over Ridge Regression, in that it produces simpler and more interpretable models that involve only a subset of the predictors.

- However, which method leads to better prediction accuracy?

- **Neither ridge regression nor the lasso will universally dominate the other**

- **A technique such as cross-validation can be used in order to determine which approach is better on a particular data set.**

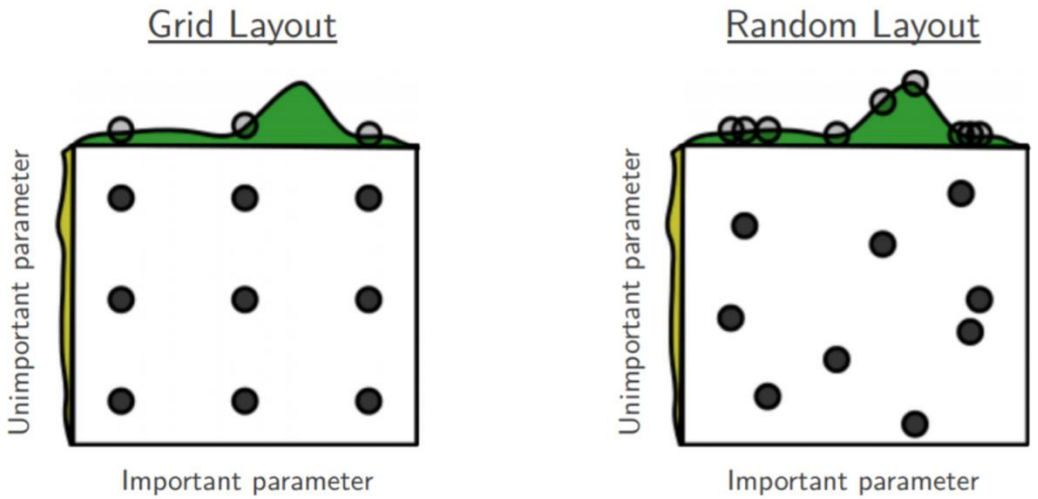
- **It is generally worth it to test both algorithms.**

5. REGULARIZATION

5.3. PARAMETER TUNING

- We have seen the value of the parameter λ is very important to ensure a good performance of the model.
- As in the case of λ, there are other important parameters to tune while we train a ML algorithm.
- Cross-validation provides a simple way to tackle this problem.
- We choose a grid of λ values, and compute the cross-validation error for each value of λ

Grid Search



Randomized Search

AGENDA

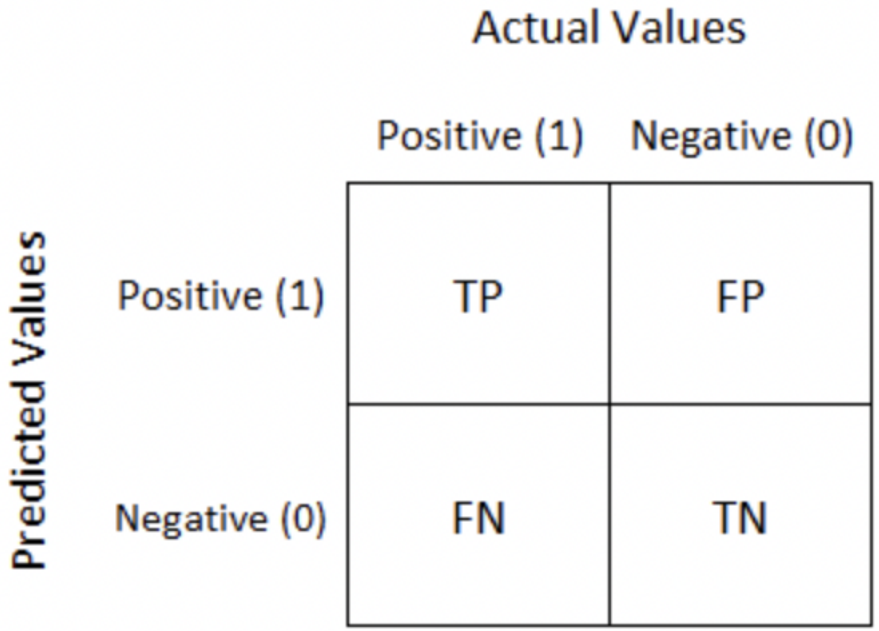
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6. EVALUATION METRICS

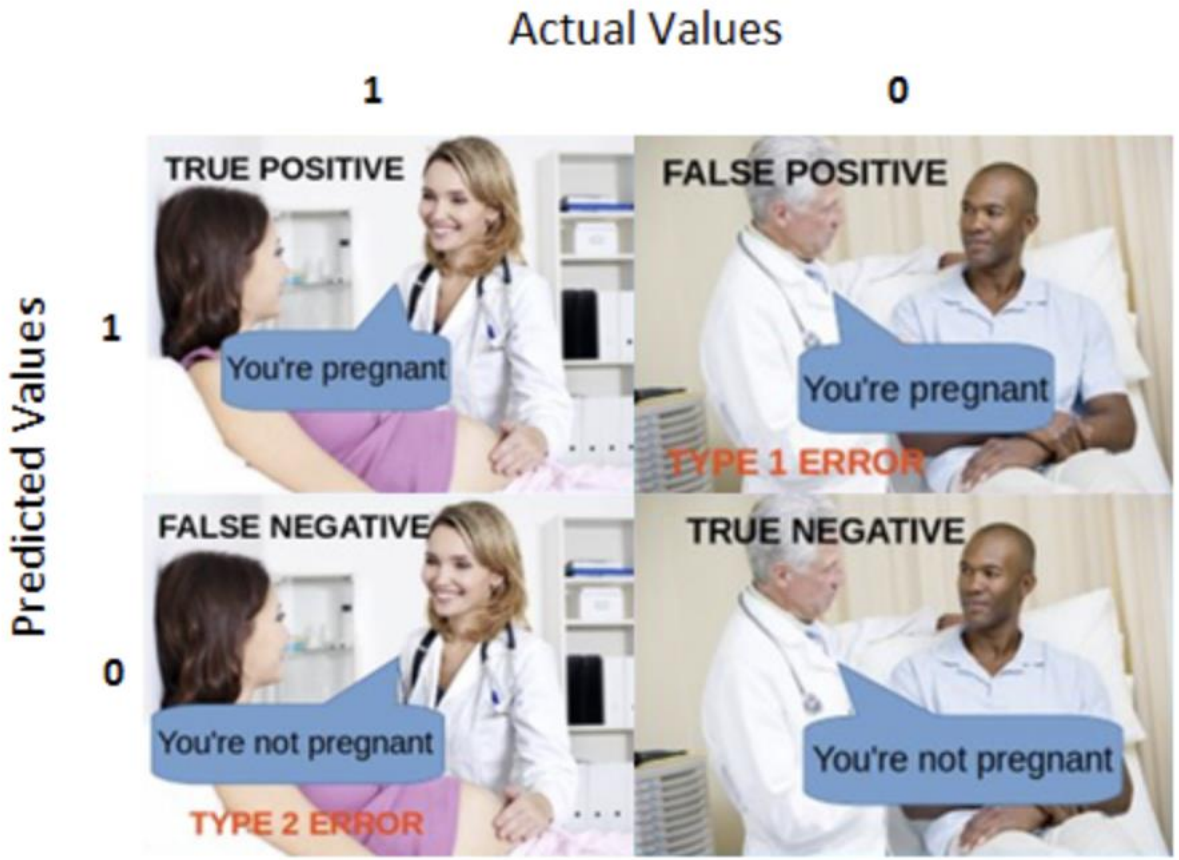
CONFUSION MATRIX

A specific table layout that allows visualization of the performance of an algorithm, typically a supervised learning one.



6. EVALUATION METRICS

CONFUSION MATRIX



6. EVALUATION METRICS

ACCURACY

Accuracy is the proportion of true results among the total number of cases examined.

Accuracy is a valid choice of evaluation for classification problems which are well balanced and not skewed or No class imbalance.

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + FP + TN + FN)}$$

PRECISION

Precision

Of all positive predictions, how many are really positive?

$$\frac{TP}{TP + FP}$$

RECALL

Recall

Of all real positive cases, how many are predicted positive?

$$\frac{TP}{TP + FN}$$

6. EVALUATION METRICS

DISCUSSION

When to use precision and recall? Give some examples.

6. EVALUATION METRICS

DISCUSSION

When to use precision and recall? Give some examples.

PRECISION: Precision is a valid choice of evaluation metric when we want to be very sure of our prediction. *For example*: *If we are building a system to predict if we should decrease the credit limit on a particular account, we want to be very sure about our prediction or it may result in customer dissatisfaction.*

RECALL: Recall is a valid choice of evaluation metric when we want to capture as many positives as possible. *For example*: *If we are building a system to predict if a person has cancer or not, we want to capture the disease even if we are not very sure.*

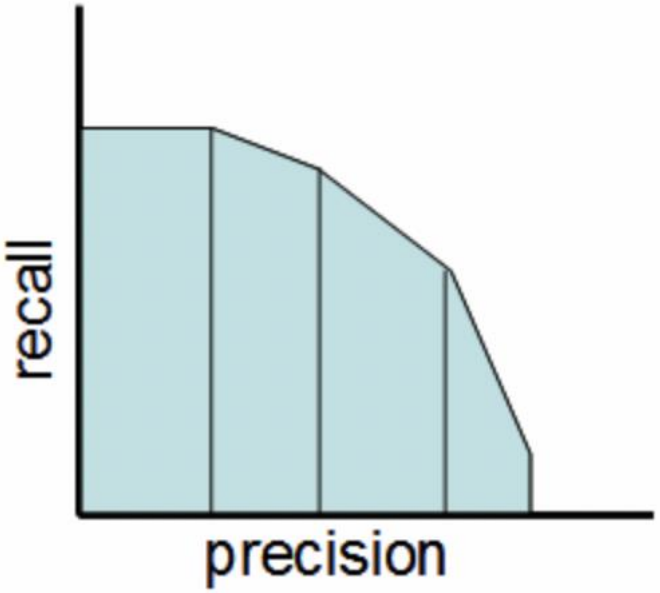
6. EVALUATION METRICS

F1 SCORE

The F1 score is a number between 0 and 1 and is the harmonic mean of precision and recall.

$$F_1 = 2 * \frac{precision * recall}{precision + recall}$$

If you are a police inspector and you want to catch criminals, you want to be sure that the person you catch is a criminal (Precision) and you also want to capture as many criminals (Recall) as possible. The F1 score manages this tradeoff.



6. EVALUATION METRICS

AUC

AUC is the area under the ROC curve. AUC ROC indicates how well the probabilities from the positive classes are separated from the negative classes. But what is the ROC curve?

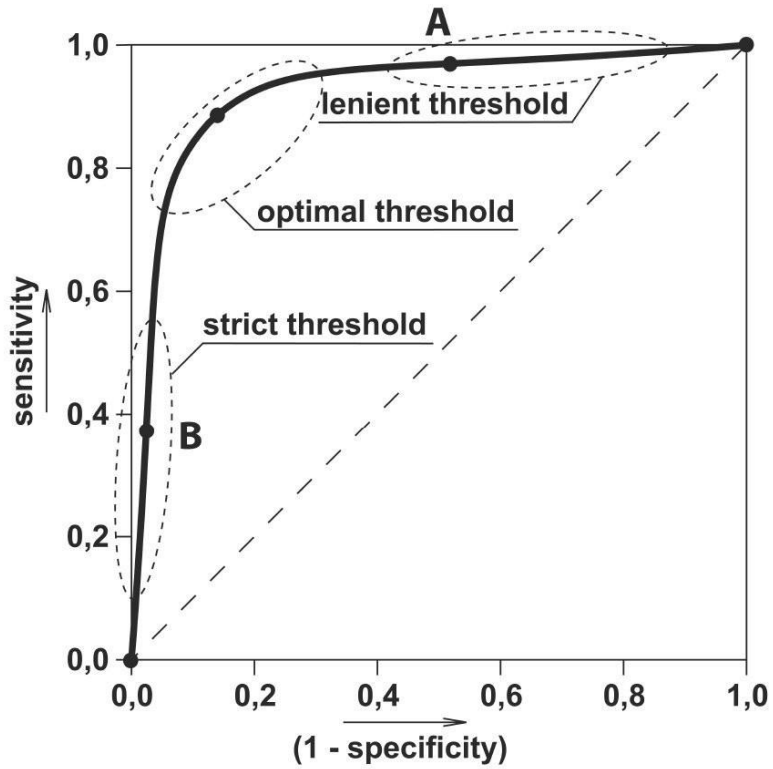
- We have got the probabilities from our classifier. We can use various threshold values to plot our **sensitivity** (TPR) and (**1-specificity**)(FPR):

- **TPR** is just the proportion of trues we are capturing.

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

- **FPR** is just the proportion of false we are capturing

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}$$




THANK YOU